

Dataset: <https://www.kaggle.com/datasets/shengkunwang/housets-dataset>

Git repo: https://github.com/jzhang2752/STAT405-Project_Group-3.git

One line code to read data: `head -1 HouseTS.csv | tr ',' '\n'` (All fields across the dataset)

Variables Descriptions:

1. Housing-market indicators (monthly):
Including variables about dynamic pricing and market activities such as sale and listing prices, inventory levels, new listings, etc.
2. Points of Interest (monthly):
Including variables about amenity based features such as restaurants, schools, parks, supermarkets etc.
3. Socioeconomic indicators (annual, temporally aligned to monthly):
Including variables about income, employment, population, education, and demographic characteristics.

Statistical methods:

1. Covid recovery
 - time-series decomposition: separate ZHVI in trend, seasonality, and noise to drag out the Covid dip and recovery
 - percent change calculation: measure each metro's drop (eg. Feb to May 2020) and recovery speed (months to return pre-covid ZHVI)
 - K-means clustering: group the 30 metros into clusters like "fast/slow recovery"
 - ANOVA: formally test whether recovery rates are statistically different across metros
2. Amenities vs. home value
 - PCA: reduce amenity types to a few key dimensions first before regression
 - correlation analysis: measure which amenity types (parks, schools, restaurants) have the strongest relationship with ZHVI levels
 - Multiple linear regression: model ZHVI growth as a function of amenity counts, controlling for other variables like income and location

Computational steps:

Data is downloaded via the Kaggle API and preprocessed through cleaning, temporal alignment, and feature construction. Jobs are parallelized by metro area on CHTC, with each node running decomposition, clustering, and regression independently before outputs are merged for final visualization.